

1 Setup and inspection

Identity / region

aws sts get-caller-identity

Confirm account and role before operating; add --profile NAME or --region REGION as needed.

List buckets

aws s3 ls

List buckets visible to your credentials.

List objects

aws s3 ls s3://BUCKET/PREFIX/ --recursive

Add --human-readable --summarize for total size/count; omit --recursive for top level.

Bucket check

aws s3api head-bucket --bucket BUCKET

Success means accessible; failure may mean missing bucket, denied access, or wrong endpoint.

Object metadata

aws s3api head-object --bucket BUCKET --key KEY

Inspect size, ETag, encryption, content type and last modified; needs GetObject.

2 Daily transfers

Upload file

aws s3 cp ./file.txt s3://BUCKET/PREFIX/

Trailing / keeps filename; specify a full destination key to rename.

Download file

aws s3 cp s3://BUCKET/PREFIX/file.txt .

The dot means current directory (PWD).

Download all

aws s3 cp s3://BUCKET/ . --recursive

Copies every object into PWD, preserving key paths. Do not add * to the S3 URI.

Upload folder

aws s3 cp ./folder/ s3://BUCKET/PREFIX/ --recursive

Copy every file under a local directory.

Sync upload

aws s3 sync ./folder/ s3://BUCKET/PREFIX/

Copy new/changed files; --delete also removes destination-only files.

Sync download

aws s3 sync s3://BUCKET/PREFIX/ ./folder/

Useful for repeat downloads; inspect source and destination direction.

S3 to S3

aws s3 cp s3://SRC/key s3://DST/key

Copy one object. Use sync for whole prefixes.

3 Useful flags and safety

Preview sync

aws s3 sync ./folder/ s3://BUCKET/PREFIX/ --dryrun

Preview actions before a large sync or any sync with --delete.

Mirror destination

aws s3 sync ./folder/ s3://BUCKET/PREFIX/ --delete

Deletes destination-only objects. Double-check source, target and account.

Filter files

aws s3 sync ./folder/ s3://BUCKET/ --exclude "*" --include "*.log"

Filter order matters: later matching rule wins. Quote wildcard patterns.

Remove object

aws s3 rm s3://BUCKET/PREFIX/file.txt

For a prefix: **aws s3 rm s3://BUCKET/PREFIX/ --recursive --dryrun** first.

Create bucket

aws s3 mb s3://BUCKET --region us-east-1

Bucket names are globally unique; pick the intended region.

Remove bucket

aws s3 rb s3://BUCKET

Requires empty bucket; versioned objects may need separate version cleanup.

4 Troubleshoot quickly

AccessDenied

Check sts identity, IAM policy, bucket policy, SCP, KMS permissions, and object ownership.

Not found / 403

Check exact bucket/key and case; without ListBucket, missing keys may return 403.

Wrong region

Use head-bucket and inspect the x-amz-bucket-region response header with --debug.

Sync surprise

Paths are bucket + prefix; run --dryrun and inspect --exclude/--include and --delete.

More detail

Append --debug for request/endpoint detail; never paste signed headers or credentials.